

## Diverse Envolving Worlds

### A Standard Simulation Worlds

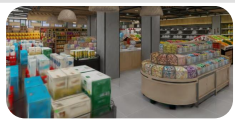


HM3D

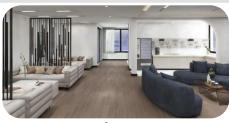


MP3D

### B Photorealistic Simulation



Habitat-GS



Interior-GS

### C Evolving Simulation Worlds



HSSD

### D Real-World Evolution



Indoor



Outdoor

Memory  
Construct

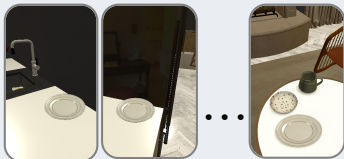


## EvolvingWorldNav



Memory  
Update

### 1 Observed History



$t_1$

$t_2$

$t_k$

### 2 Hidden Evolution



world evolves

### 3 Current Belief



state<sub>1</sub>

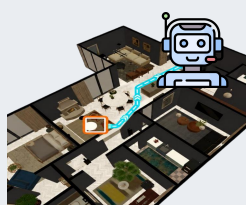


state<sub>2</sub>



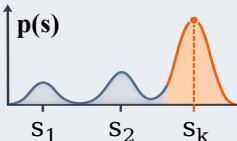
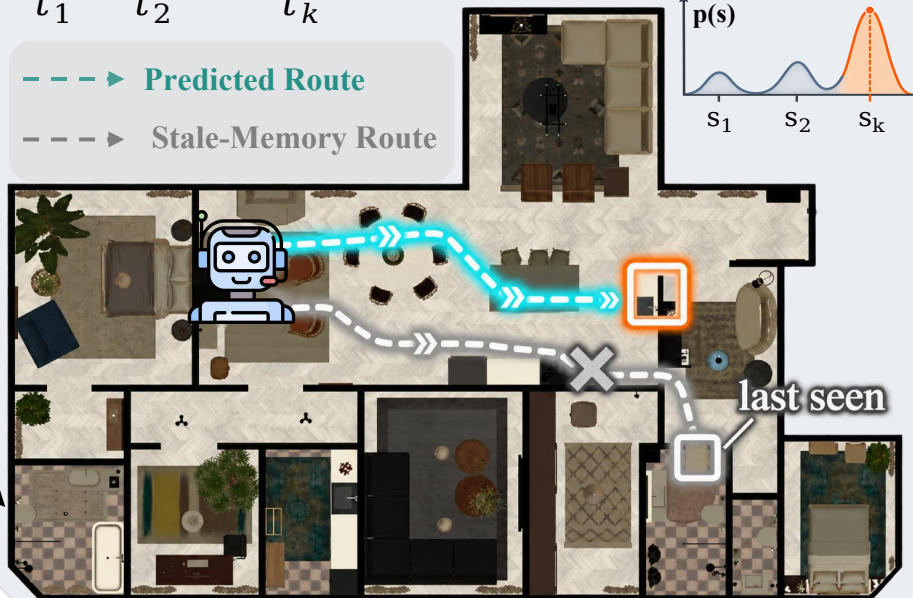
state<sub>k</sub>

### 4 Belief-guided Plan/Act



---> Predicted Route

---> Stale-Memory Route



### P4D Bench



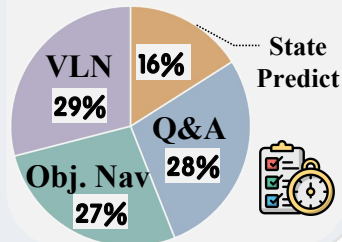
54 Scenes

803.68k Tasks

90 Scene-Days

239.19k Dynamic

State Changes



## Unified Embodied Tasks

### A Embodied QA

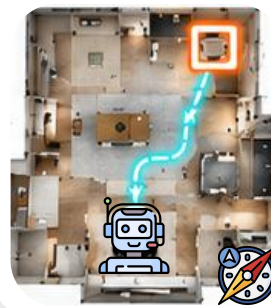


Q: Where did I put the cup after I left the dining table ?

### B Active Exploration



### C Predictive Navigation



### D Real World Interaction

